화학컴퓨터 (2009년 1학기)

1차 중간고사 정답

(**총 11 문제, 150점**)

1. **(총10점, 각5점)**

(a) Euler 공식을 이용하면,

$$\begin{aligned} e^{i(3\pi/4)+\ln 2} &= e^{\ln 2}e^{i(3\pi/4)} = 2\left(\cos\left(\frac{3\pi}{4}\right)+i\sin\left(\frac{3\pi}{4}\right)\right) \\ &= 2\left(-\frac{\sqrt{2}}{2}+\frac{\sqrt{2}}{2}i\right) = -\sqrt{2}+\sqrt{2}i. \end{aligned}$$

그러므로 실수부는 $-\sqrt{2}$이고 허수부는 $\sqrt{2}$이다.

(b) $1+\sqrt{3}i = 2\left(\frac{1}{2}+\frac{\sqrt{3}}{2}i\right) = 2\big(\cos(\pi/3)+i\sin(\pi/3)\big) = 2e^{i\pi/3}$이므로,

$$\begin{aligned} \sqrt{1+\sqrt{3}i}+2e^{i\pi/3} &= \left(2e^{i\pi/3}\right)^{1/2}+2e^{i\pi/3} \\ &= \sqrt{2}e^{i\pi/6}+2e^{i\pi/3} \\ &= \sqrt{2}\left(\cos\left(\frac{\pi}{6}\right)+i\sin\left(\frac{\pi}{6}\right)\right)+2\left(\cos\left(\frac{\pi}{3}\right)+i\sin\left(\frac{\pi}{3}\right)\right) \\ &= \sqrt{2}\left(\frac{\sqrt{3}}{2}+\frac{1}{2}i\right)+2\left(\frac{1}{2}+\frac{\sqrt{3}}{2}i\right) \\ &= \left(\frac{\sqrt{6}}{2}+1\right)+\left(\frac{\sqrt{2}}{2}+\sqrt{3}\right)i. \end{aligned}$$

그러므로 실수부분은 $\sqrt{6}/2+1$이고 허수부분은 $\sqrt{2}/2+\sqrt{3}$이다.

2. **(총15점, 각5점)**

(a) $f(-x) = (e^{-x}-e^{x})/2 = -(e^{x}-e^{-x})/2 = -f(x)$이므로 **기함수**이다.

(b) $f(-x) = e^{-x}\sin(-x) = -e^{-x}\sin x$이므로 **우함수나 기함수가 아니다**.

(c)

$$\begin{aligned} f(-x) &= \frac{e^{-x}}{(e^{-x}+1)^2} \\ &= \frac{e^{-x}}{(e^{-x}+1)^2}\times\frac{e^{2x}}{e^{2x}} \\ &= \frac{e^{x}}{(1+e^{x})^2} = f(x). \end{aligned}$$

그러므로 **우함수**이다.

3. **(총10점, 각5점)**

(a) $y = (2+x)e^{-x^2}$

$$\begin{aligned} \frac{dy}{dx} &= e^{-x^2}+(2+x)(-2x)e^{-x^2} = (1-4x-2x^2)e^{-x^2} \\ \frac{d^2y}{dx^2} &= (-4-4x)e^{-x^2}+(1-4x-2x^2)(-2x)e^{-x^2} = (-4-6x+8x^2+4x^3)e^{-x^2}. \end{aligned}$$

(b) $y = e^{-x}\sin x$

$$\begin{aligned}\frac{dy}{dx} &= -e^{-x}\sin x + e^{-x}\cos x = (-\sin x + \cos x)e^{-x} \\ \frac{d^2y}{dx^2} &= (-\cos x - \sin x)e^{-x} + (-\sin x + \cos x)(-1)e^{-x} = -2\cos x e^{-x}.\end{aligned}$$

4. **(총 10점, 각 5점)**

(a) $f(x,y) = \ln(ax^2 + by^2)$ $\quad$ (a, b는 상수.)

$$\begin{aligned}\left(\frac{\partial f}{\partial x}\right)_y &= \frac{2ax}{ax^2+by^2}, \\ \left(\frac{\partial f}{\partial y}\right)_x &= \frac{2by}{ax^2+by^2}, \\ \left(\frac{\partial^2 f}{\partial x^2}\right)_y &= \frac{2a(ax^2+by^2) - 2ax2ax}{(ax^2+by^2)^2} = \frac{-2a^2x^2+2aby^2}{(ax^2+by^2)}, \\ \left(\frac{\partial^2 f}{\partial y^2}\right)_x &= \frac{2b(ax^2+by^2) - 2by2by}{(ax^2+by^2)^2} = \frac{2abx^2-2b^2y^2}{(ax^2+by^2)^2}, \\ \left(\frac{\partial^2 f}{\partial x\partial y}\right) &= \frac{-2by2ax}{(ax^2+by^2)^2} = \frac{-4abxy}{(ax^2+by^2)^2}, \\ \left(\frac{\partial^2 f}{\partial y\partial x}\right) &= \frac{-2ax2by}{(ax^2+by^2)^2} = \frac{-4abxy}{(ax^2+by^2)^2},\end{aligned}$$

(b) $f(x,y) = \cos(x/y)$

$$\begin{aligned}\left(\frac{\partial f}{\partial x}\right)_y &= -\frac{1}{y}\sin\left(\frac{x}{y}\right) \\ \left(\frac{\partial f}{\partial y}\right)_x &= -\sin\left(\frac{x}{y}\right)\cdot -\frac{x}{y^2} = \frac{x}{y^2}\sin\left(\frac{x}{y}\right) \\ \left(\frac{\partial^2 f}{\partial x^2}\right)_y &= \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial x}\right)_y = \frac{\partial}{\partial x}\left[-\frac{1}{y}\sin\left(\frac{x}{y}\right)\right] = -\frac{1}{y^2}\cos\left(\frac{x}{y}\right) \\ \left(\frac{\partial^2 f}{\partial y^2}\right)_x &= \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial y}\right)_x = \frac{\partial}{\partial y}\left[\frac{x}{y^2}\sin\left(\frac{x}{y}\right)\right] = -\frac{2x}{y^3}\sin\left(\frac{x}{y}\right) + \frac{x}{y^2}\cos\left(\frac{x}{y}\right)\cdot -\frac{x}{y^2} \\ &= -\frac{2x}{y^3}\sin\left(\frac{x}{y}\right) - \frac{x^2}{y^4}\cos\left(\frac{x}{y}\right) \\ \left(\frac{\partial^2 f}{\partial x\partial y}\right) &= \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial y}\right)_x = \frac{\partial}{\partial x}\left[\frac{x}{y^2}\sin\left(\frac{x}{y}\right)\right] = \frac{1}{y^2}\sin\left(\frac{x}{y}\right) + \frac{x}{y^3}\cos\left(\frac{x}{y}\right) \\ \left(\frac{\partial^2 f}{\partial y\partial x}\right) &= \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial x}\right)_y = \frac{\partial}{\partial y}\left[-\frac{1}{y}\sin\left(\frac{x}{y}\right)\right] = \frac{1}{y^2}\sin\left(\frac{x}{y}\right) + \frac{x}{y^3}\cos\left(\frac{x}{y}\right)\end{aligned}$$

5. **(총 10점, 각 5점)**

(a) l'Hôpital의 정리를 두 번 이용하면

$$\lim_{x\to 0}\frac{x^2e^{-x}}{(1-e^{-x})^2} = \lim_{x\to 0}\frac{2xe^{-x}-x^2e^{-x}}{2(1-e^{-x})e^{-x}} = \lim_{x\to 0}\frac{2x-x^2}{2(1-e^{-x})} = \lim_{x\to 0}\frac{2-2x}{2e^{-x}} = 1.$$

(b) l'Hôpital의 정리를 두 번 이용하면

$$\lim_{x\to 0}\frac{\ln(1+x)-x}{x^2}=\lim_{x\to 0}\frac{\frac{1}{1+x}-1}{2x}==\lim_{x\to 0}\frac{-\frac{1}{(1+x)^2}}{2}=-\frac{1}{2}$$

6. **(총 15 점, 각 5 점)**

(a) $d\bar{S}=\frac{\bar{C}_V}{T}dT+\frac{R}{\bar{V}}d\bar{V}\Rightarrow$ **완전미분**, $\quad \bar{S}=\bar{C}_V\ln T+R\ln\bar{V}+C.$

(b) $dP=\frac{RT}{\bar{V}-b}dT+\left[\frac{RT}{(\bar{V}-b)^2}-\frac{a}{T\bar{V}^2}\right]d\bar{V}\Rightarrow$ **불완전미분**

(c)

$$\begin{aligned} dP &= \left[\frac{R}{\bar{V}-B}+\frac{A}{2T^{3/2}\bar{V}(\bar{V}+B)}\right]dT+\left[-\frac{RT}{(\bar{V}-B)^2}+\frac{A(2\bar{V}+B)}{T^{1/2}\bar{V}^2(\bar{V}+B)^2}\right]d\bar{V} \\ &\Rightarrow \quad \textbf{완전미분}, \qquad P=\frac{RT}{\bar{V}-B}-\frac{A}{2T^{1/2}\bar{V}(\bar{V}+B)}+C. \end{aligned}$$

7. **(총 10 점, 각 5 점)**

(a) $\int x\ln x dx$ 만약 $u=\ln x$ 그리고 $v'=xdx$ 라면, $u'=\frac{dx}{x}$ 그리고 $v=\frac{1}{2}x^2$ 이므로,

$$\begin{aligned} \int x\ln x dx &= \frac{1}{2}x^2\ln x-\int\frac{1}{2}x^2\frac{1}{x}dx \\ &= \frac{1}{2}x^2\ln x-\frac{1}{2}\int xdx \\ &= \frac{1}{2}x^2\ln x-\frac{1}{4}x^2+C. \end{aligned}$$

(b) 만약 $t=x^2+1$ 로 치환하면, $dt=2xdx$ 이므로

$$\begin{aligned} \int\frac{x^3}{1+x^2}dx &= \int\frac{x^2xdx}{1+x^2} \\ &= \int\frac{(t-1)\frac{1}{2}dt}{t} \\ &= \frac{1}{2}\int(1-\frac{1}{t})dt \\ &= \frac{1}{2}\Big(t-\ln t\Big) \\ &= \frac{1}{2}\big(x^2+1-\ln(x^2+1)\big) \end{aligned}$$

8. **(총 20 점, 각 5 점)**

(a)

$$\int_0^a x(a-x)\sin\left(\frac{n\pi x}{a}\right)dx \quad = \quad \int_0^a\big(ax-x^2\big)\sin\left(\frac{n\pi x}{a}\right)dx. \qquad (1)$$

(1) 식에서 $u=ax-x^2$ 그리고 $v'=\sin\Big(\frac{n\pi x}{a}\Big)dx$ 라면, $u'=(a-2x)dx$ 그리고

$v = -\frac{a}{n\pi}\cos\left(\frac{n\pi x}{a}\right)$ 이므로 (1) 식은 다음과 같이 쓸 수 있다.

$$\begin{aligned}\int_0^a x(a-x)\sin\left(\frac{n\pi x}{a}\right)dx &= \int_0^a \left(ax - x^2\right)\sin\left(\frac{\pi x}{a}\right)dx, \\ &= \left[-\frac{a}{n\pi}(ax-x^2)\cos\left(\frac{n\pi x}{a}\right)\right]_0^a + \frac{a}{n\pi}\int_0^a (a-2x)\cos\left(\frac{n\pi x}{a}\right)dx \\ &= \frac{a}{n\pi}\int_0^a (a-2x)\cos\left(\frac{n\pi x}{a}\right)dx. \qquad (2)\end{aligned}$$

(2) 식에서 $u = a - 2x$ 그리고 $v' = \cos\left(\frac{n\pi x}{a}\right)dx$ 라면, $u' = -2dx$ 그리고 $v = \frac{a}{n\pi}\sin\left(\frac{n\pi x}{a}\right)$ 이므로 (2) 식은 다음과 같이 쓸 수 있다.

$$\begin{aligned}\frac{a}{n\pi}\int_0^a (a-2x)\cos\left(\frac{n\pi x}{a}\right)dx &= \left[\left(\frac{a}{n\pi}\right)^2 (a-2x)\sin\left(\frac{n\pi x}{a}\right)\right]_0^a + 2\left(\frac{a}{n\pi}\right)^2\int_0^a \sin\left(\frac{n\pi x}{a}\right)dx \\ &= \left(\frac{a}{n\pi}\right)^2(-a)\sin(n\pi) + 2\left(\frac{a}{n\pi}\right)^2\int_0^a \sin\left(\frac{n\pi x}{a}\right)dx \\ &= 2\left(\frac{a}{n\pi}\right)^2\int_0^a \sin\left(\frac{n\pi x}{a}\right)dx \qquad (\sin(n\pi) = 0), \\ &= -2\left(\frac{a}{n\pi}\right)^3\left[\cos\left(\frac{n\pi x}{a}\right)\right]_0^a \\ &= 2\left(\frac{a}{n\pi}\right)^3\left(1-\cos(n\pi)\right) \\ &= 2\left(\frac{a}{n\pi}\right)^3\left(1-(-1)^n\right). \qquad (3)\end{aligned}$$

(b)

$$\begin{aligned}\int_{T_1}^{T_2}\left(\frac{a}{T} + b + cT + dT^2\right)dT &= \left[a\ln T + bT + \frac{c}{2}T^2 + \frac{d}{3}T^3\right]_{T_1}^{T_2}, \\ &= a(\ln T_2 - \ln T_1) + b(T_2 - T_1) + \frac{c}{2}(T_2^2 - T_1^2) + d\frac{d}{3}(T_2^3 - T_1^3).\end{aligned}$$

(c) $t = J^2 + J$ 로 치환하면 , $dt = (2J+1)dJ$ 이 되므로

$$\int_0^\infty (2J+1)e^{-a(J^2+J)}dJ = \int_0^\infty e^{-at}dt = \frac{1}{a}$$

(d)

$$\frac{1}{16}\int_0^\infty\int_0^\pi r^4 e^{-r}\sin^3\theta dr d\theta = \frac{1}{16}\underbrace{\int_0^\infty r^4 e^{-r}dr}_{(A)}\underbrace{\int_0^\pi \sin^3\theta d\theta}_{(B)}.$$

부분적분법을 이용하여 (A) 를 적분하기 위하여 $u = r^4$ 그리고 $v' = e^{-r}$ 라면, $u' =$

$4r^3dr,\ v=-e^{-r}$ 이므로,

$$\begin{aligned}(A)=\int_0^\infty r^4e^{-r}dr &= \Big[-r^4e^{-r}\Big]_0^\infty+\int_0^\infty 4r^3e^{-r}dr,\\ &= 4\int_0^\infty r^3e^{-r}dr, \qquad (4)\end{aligned}$$

다시 부분적분법을 이용하여 (4) 식을 적분하기 위하여 $u=r^3$ 그리고 $v'=e^{-r}$ 라면, $u'=3r^2dr,\ v=-e^{-r}$ 이므로,

$$\begin{aligned}(A)=\int_0^\infty r^4e^{-r}dr &= 4\int_0^\infty r^3e^{-r}dr,\\ &= 4\Big[-r^3e^{-r}\Big]_0^\infty+12\int_0^\infty r^2e^{-r}dr,\\ &= 12\int_0^\infty r^2e^{-r}dr, \qquad (5)\end{aligned}$$

다시 부분적분법을 이용하여 (4) 식을 적분하기 위하여 $u=r^2$ 그리고 $v'=e^{-r}$ 라면, $u'=2rdr,\ v=-e^{-r}$ 이므로,

$$\begin{aligned}(A)=\int_0^\infty r^4e^{-r}dr &= 12\int_0^\infty r^2e^{-r}dr,\\ &= 12\Big[-r^2e^{-r}\Big]_0^\infty+24\int_0^\infty re^{-r}dr,\\ &= 24\int_0^\infty re^{-r}dr, \qquad (6)\end{aligned}$$

다시 부분적분법을 이용하여 (6) 식을 적분하기 위하여 $u=r$ 그리고 $v'=e^{-r}$ 라면, $u'=dr,\ v=-e^{-r}$ 이므로,

$$\begin{aligned}(A)=\int_0^\infty r^4e^{-r}dr &= 12\int_0^\infty r^2e^{-r}dr,\\ &= 24\Big[-re^{-r}\Big]_0^\infty+24\int_0^\infty e^{-r}dr,\\ &= 24\int_0^\infty e^{-r}dr,\\ &= 24\Big[-e^{-r}\Big]_0^\infty=24. \qquad (7)\end{aligned}$$

또는, $\int_0^\infty x^ne^{-ax}dx=\frac{n!}{a^{n+1}}$ 을 이용하면,

$$(A)=\int_0^\infty r^4e^{-r}dr=4!=24.$$

한편, (B) 에서 $\sin^3\theta=\sin^2\theta\cdot\sin\theta=(1-\cos^2\theta)\sin\theta$ 이므로 $\cos\theta=t$ 로 치환하면

$-\sin\theta d\theta = dt$ 가 되고,

$$\begin{aligned}(B) = \int_0^{\pi} \sin^3\theta d\theta &= \int_0^{\pi} \sin^2\theta \cdot \sin\theta\theta d\theta = \int_0^{\pi} (1-\cos^2\theta)\sin\theta d\theta \\ &= \int_1^{-1} (1-t^2)(-dt) \\ &= \int_{-1}^{1} (1-t^2)dt \\ &= \left[t - \frac{1}{3}t^3\right]_{-1}^{1} \\ &= \left(1-\frac{1}{3}\right) - \left(-1+\frac{1}{3}\right) = \frac{4}{3}. \end{aligned} \tag{8}$$

최종적으로

$$\begin{aligned}\frac{1}{16}\int_0^{\infty}\int_0^{\pi} r^4 e^{-r}\sin^3\theta dr d\theta &= \frac{1}{16}\underbrace{\int_0^{\infty} r^4 e^{-r} dr}_{(A)}\underbrace{\int_0^{\pi}\sin^3\theta d\theta}_{(B)} \\ &= \frac{1}{16}\times 24 \times \frac{4}{3} = 2. \end{aligned}$$

9. **(총 20점, 각 5점)**

(a)

$$\int_C df(x,y) = \int_C (2xydx + x^2dy), \tag{9}$$

위 식에서 경로 C는 점 $(1,1)$부터 점 $(5,5)$을 연결하는 직선 $y-x=0$을 의미한다.

$$\begin{aligned}\int_C df(x,y) &= \int_1^5 2x^2dx + \int_1^5 y^2dy \\ &= \left[\frac{2}{3}x^3\right]_1^5 + \left[\frac{1}{3}y^3\right]_1^5 = 124. \end{aligned} \tag{10}$$

(b)

$$\int_C df(x,y) = \int_C (2xydx + x^2dy), \tag{11}$$

위 식에서 경로 C는 처음에는 $(1,1)$에서 $(5,1)$까지 그 다음에는 $(5,1)$에서 $(5,5)$로 주어진다.

$$\begin{aligned}\int_C df(x,y) &= \int_1^5 2x\times 1dx + \int_1^5 5^2dy \\ &= \left[x^2\right]_1^5 + \left[25y\right]_1^5 = 124. \end{aligned} \tag{12}$$

(c) $M(x,y) = 2xy,\ N(x,y) = x^2$ 이므로

$$\left(\frac{\partial M}{\partial y}\right)_x = 2x, \qquad \left(\frac{\partial N}{\partial x}\right)_y = 2x \Rightarrow \left(\frac{\partial M}{\partial y}\right)_x = \left(\frac{\partial N}{\partial x}\right)_y$$

이므로, $df(x,y) = 2xydx + x^2dy$ 는 완전미분이다.

(d)

$$\begin{aligned}\int_C df(x,y) &= \int_{(1,1)}^{(5,5)} (2xydx + x^2dy) \\ &= \Big[x^2y\Big]_{(1,1)}^{(5,5)} = 124. \end{aligned} \tag{13}$$

10. **(10점)**

$$N_J = a(2J+1)e^{-BJ(J+1)/k_BT} \qquad (a \text{ 는 비례상수}).$$

이므로,

$$\begin{aligned}\frac{dN_J}{dJ} &= 2ae^{-BJ(J+1)/k_BT} - a\frac{B}{k_BT}(2J+1)^2 e^{-BJ(J+1)/k_BT} \\ &= \left(2 - \frac{B}{k_BT}(2J+1)^2\right) ae^{-BJ(J+1)/k_BT}\end{aligned}$$

N_J 가 최대값일 때, $dN_J/dJ = 0$ 이 되고 그 때의 $J = J_{max}$ 는 다음과 같이 구할 수 있다.

$$\left(2 - \frac{B}{k_BT}(2J_{max}+1)^2\right) ae^{-BJ_{max}(J+1_{max})/k_BT} \Rightarrow 2 - \frac{B}{k_BT}(2J_{max}+1)^2 = 0$$
$$\Rightarrow J_{max} = -\frac{1}{2} \pm \frac{1}{2}\sqrt{\frac{2k_BT}{B}}.$$

$J_{max} > 0$ 이므로 $J_{max} = -1/2 + \sqrt{k_BT/2B}$.

11. **(총 20점, 각 5점)**

(a)

$$dU = \Big(\frac{\partial U}{\partial T}\Big)_V dT + \Big(\frac{\partial U}{\partial V}\Big)_T dV. \tag{14}$$

$$dS = \Big(\frac{\partial S}{\partial T}\Big)_V dT + \Big(\frac{\partial S}{\partial V}\Big)_T dV. \tag{15}$$

(b)

$$\begin{aligned} dU &= TdS - PdV \\ &= T\left[\Big(\frac{\partial S}{\partial T}\Big)_V dT + \Big(\frac{\partial S}{\partial V}\Big)_T dV\right] - PdV \qquad ((15) \text{로부터}) \\ &= T\Big(\frac{\partial S}{\partial T}\Big)_V dT + \left[T\Big(\frac{\partial S}{\partial V}\Big)_T - P\right] dV \end{aligned} \tag{16}$$

(14) 식과 (16) 식을 비교하면

$$\boxed{\left(\frac{\partial U}{\partial V}\right)_T = T\left(\frac{\partial S}{\partial V}\right)_T - P.} \tag{17}$$

(c) Helmholtz 자유에너지의 미분형태 $dA = -SdT - PdV$ 와 Maxwell 관계식을 이용하면,

$$\left(\frac{\partial S}{\partial V}\right)_T = \left(\frac{\partial P}{\partial T}\right)_V$$

이므로, (17) 식은 다음과 같이 쓸 수 있다.

$$\boxed{\left(\frac{\partial U}{\partial V}\right)_T = T\left(\frac{\partial S}{\partial V}\right)_T - P = T\left(\frac{\partial P}{\partial T}\right)_V - P.} \tag{18}$$

(d) 이상기체의 경우에 $PV = nRT$ 이므로 $P = nRT/V$ 이므로,

$$\left(\frac{\partial P}{\partial T}\right)_V = \frac{nR}{V} \tag{19}$$

(19) 식을 (17) 식에 대입하면

$$\boxed{\left(\frac{\partial U}{\partial V}\right)_T = T\left(\frac{\partial P}{\partial T}\right)_V - P = \frac{nRT}{V} - P = \frac{nRT}{V} - \frac{nRT}{V} = 0.}$$

van der Waals 기체의 경우에

$$P = \frac{nRT}{V - nb} - \frac{n^2 a}{V^2} \tag{20}$$

이므로,

$$\left(\frac{\partial P}{\partial T}\right)_V = \frac{nR}{V - nb}. \tag{21}$$

(21) 식을 (17) 식에 대입하면

$$\boxed{\left(\frac{\partial U}{\partial V}\right)_T = T\left(\frac{\partial P}{\partial T}\right)_V - P = \frac{nRT}{V - nb} - P = \frac{nRT}{V} - \frac{nRT}{V - nb} + \frac{n^2 a}{V^2} = \frac{n^2 a}{V^2}.}$$